Nikunj Pradhan

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Education

Missouri University of Science and Technology GPA: 4.0/4.0

Masters of Computer Science *Aug 2025 - May 2027*

Bachelor of Science: Computer Science *Aug 2023 - Dec 2026*

Experience

Data Science Co-Op Jan 2025 – Aug 2025

Hunter Engineering Company *Bridgeton, MO*

- Improved end-to-end object detection/classification pipeline accuracy from 68% to 84% by benchmarking multiple YOLO architectures and optimizing hyperparameters.
- Developed a ResNet-based CNN regression model to estimate four corner coordinates of distorted quadrilateral regions for perspective rectification, reducing geometric correction error by 23%.
- Created and maintained custom ETL pipelines to process metadata and image data from 600+ worksites, ingesting over 2M images into Microsoft SQL and AWS S3.
- Built automated benchmarking pipelines across cloud platforms with serverless functions (Lambda, Azure Functions) to compare performance of in-house and third-party AI/ML models.
- Reduced model training time by 66% by migrating vision models to PyTorch Lightning and enabling multi-GPU training.

Software Engineer June 2025 – October 2025

Tally Receipts *St. Louis, MO*

- Redesigned and migrated backend from MongoDB to AWS DynamoDB, redesigning schemas for efficient querying, scalability, and data lifecycle management.
- Built an LLM-powered receipt management system with secure card-linking for transaction ingestion, with webhook-driven backend pipelines and calendar sync, using LlamaIndex for receipt summarization and mapping transactions to calendar events.

Research Assistant Aug 2023 – Dec 2024

Missouri University of Science and Technology *Rolla, MO*

- Optimized algorithms, increasing blob analysis accuracy 62% through hyperparameter tuning, and algorithmic refinement using C++ OpenCV and MATLAB.
- Designed a C++/MATLAB application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling with Point Cloud Library.
- Utilized watershed segmentation and 3D model triangulation via ray tracing, and used a KNN-based algorithm for point classification.

Projects

Nepali Devanagari OCR Pipeline | *PyTorch, Python, FastAPI, NLTK, Docker, ONNX*

- Designed a CRNN-based Nepali OCR pipeline covering 130+ characters, combining CNN backbones with LSTM/Bi-LSTM sequence modeling and utilized CTC and Additive Attention decoding mechanisms.
- Collected, annotated, and augmented 1000+ images into 500,000+ training samples and utilized the model via a FastAPI inference service for real-time OCR for text recognition and line detection.

Hybrid GNN-Mamba Model for Toxicity Prediction | *PyTorch Geometric, Mamba-SSM, RDKit*

- Designed a parallel GIN + Mamba hybrid architecture for multi-task molecular toxicity prediction on Tox21. Innovated a graph-to-sequence mapping strategy achieving 75% ROC-AUC and 36% PR-AUC under scaffold split

Movie Recommendation & Watchlist Platform | *FastAPI, React, PostgreSQL, Scikit-learn, K3S, Docker*

- Deployed a full-stack movie watchlist platform with personalized recommendations using TF-IDF and cosine similarity for content-based filtering alongside ALS collaborative filtering, with per-user caching to reduce latency.

Technical Skills

Programming Languages: Python, C++, JavaScript, SQL, Java, HTML/CSS
Machine Learning Tools: Pytorch, Scikit-learn, LangChain, LlamaIndex, XGBoost, Transformers
Web Development & APIs: FastAPI, Spring Boot, React, Node.js, Apache Kafka, WebSockets
Libraries & Tools : Numpy, Pandas, NLTK, Matplotlib, OpenCV, SciPy, Redis, Playwright, BeautifulSoup
Databases: PostgreSQL, Pgvector, Microsoft SQL, DynamoDB, MinIO, MongoDB
Systems & DevOps: AWS, Docker, Kubernetes (k8s), Git, CI/CD, Ubuntu, Jupyter, CUDA